

A WAP Pullback Reduction for Hanson Question 4.3

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Status

This is a conditional reduction, not a full solution. It isolates a functional-analytic lemma which would imply a positive answer to Question 4.3 of Hanson [1]. The model-theoretic part of the reduction is proved below; the Banach-Ramsey lemma is left as an explicit dependency.

Source Question

Hanson asks:

Does (model theoretic) stability imply oscillation stability? That is to say, if T is a stable Banach theory, is every unary formula oscillation stable on models of T ?

Question 4.3. *Does (model theoretic) stability imply oscillation stability? That is to say, if T is a stable Banach theory, is every unary formula oscillation stable on models of T ?*

4.2. The (Type-)Definability of Indiscernible Subspaces and Complex Banach Structures A central question in the study of inseparably categorical Ba-

Definitions

Let X be an infinite-dimensional Banach space and let S_X denote its unit sphere. A function $f : S_X \rightarrow \mathbb{R}$ is *oscillation stable* if for every infinite-dimensional subspace $Y \subseteq X$ and every $\varepsilon > 0$, there is an infinite-dimensional subspace $Z \subseteq Y$ such that

$$|f(z) - f(z')| \leq \varepsilon \quad (z, z' \in S_Z).$$

The reduction uses the following analytic stability property.

Definition 1 (Midpoint-WAP stability). *Let $F : B_X \rightarrow \mathbb{R}$ be bounded and uniformly continuous. Say that F is midpoint-WAP if the kernel*

$$K_F(u, v) = F\left(\frac{u+v}{2}\right), \quad u, v \in B_X,$$

has the Grothendieck double-limit property: whenever (u_n) and (v_m) are sequences in B_X and both iterated limits exist,

$$\lim_n \lim_m K_F(u_n, v_m) = \lim_m \lim_n K_F(u_n, v_m).$$

By Grothendieck's criterion, in the formulation recalled by Ben Yaacov [2], this is the weak almost periodic/relative weak compactness condition for the midpoint-translation family generated by F .

Conditional Dependencies

Conditional Dependency 1 (WAP midpoint oscillation lemma). *Let X be an infinite-dimensional Banach space. Every bounded uniformly continuous midpoint-WAP function $F : B_X \rightarrow \mathbb{R}$ is oscillation stable on S_X .*

This is the only unproved dependency in the packet. It is a pure Banach-space statement: it mentions neither formulas nor elementary extensions. It should be attacked directly with weakly almost periodic/reflexive representation tools and block Ramsey methods, or refuted by constructing a midpoint-WAP version of Maurey’s non-oscillation-stable Hilbert-sphere coloring.

Idea of the Proof

Hanson’s Dvoretzky–Milman compactness argument shows that finite-dimensional almost-monochromatic subspaces are always available, even without stability; the obstruction is pulling the resulting ultraproduct subspace back into the original model. Stability changes the unary formula instance $x \mapsto \varphi(x, a)$ into a weakly almost periodic object after midpoint translation, because $(u, v) \mapsto \varphi((u + v)/2, a)$ is again a formula and therefore stable. Thus the remaining missing bridge is exactly whether midpoint-WAP functions have the infinitary oscillation-stability property.

Conditional Theorem

Theorem 1. *Assume Conjecture 1. Then Hanson Question 4.3 has a positive answer: if T is a stable Banach theory, $M \models T$, and $\varphi(x, a)$ is any unary formula instance on the unit ball of M , then the function $x \mapsto \varphi^M(x, a)$ is oscillation stable on S_M .*

Proof. Fix $M \models T$, parameters $a \in M$, and a unary formula instance $\varphi(x, a)$. Let

$$F(x) = \varphi^M(x, a), \quad x \in B_M.$$

The function F is bounded and uniformly continuous because formulas in continuous logic are bounded and uniformly continuous on each sort.

Define a two-variable formula

$$\theta(u, v, a) = \varphi\left(\frac{u + v}{2}, a\right).$$

The midpoint operation is part of the Banach structure, so θ is a formula. Since T is stable, every formula is stable in every model of T . Hence for all sequences $(u_n), (v_m)$ in B_M , whenever both iterated limits exist,

$$\lim_n \lim_m \theta^M(u_n, v_m, a) = \lim_m \lim_n \theta^M(u_n, v_m, a).$$

Equivalently, F is midpoint-WAP on B_M .

Now let $Y \subseteq M$ be any infinite-dimensional closed subspace. The restriction $F|_{B_Y}$ is still bounded, uniformly continuous, and midpoint-WAP, because the double-limit condition is inherited by restricting the allowed sequences to B_Y . Applying Conjecture 1 to Y , for every $\varepsilon > 0$ there exists an infinite-dimensional subspace $Z \subseteq Y$ such that

$$|\varphi^M(z, a) - \varphi^M(z', a)| \leq \varepsilon \quad (z, z' \in S_Z).$$

This is exactly oscillation stability of the unary formula instance $\varphi(x, a)$. Since M, a, φ, Y , and ε were arbitrary, the conclusion follows. $\square$

Verification Notes

The proof of Theorem 1 uses only three standard facts:

1. Banach continuous logic includes affine combinations such as $(u + v)/2$ as uniformly continuous structure maps between ball sorts.
2. In a stable theory, every formula, including $\theta(u, v, a) = \varphi((u + v)/2, a)$, has the double-limit property.
3. Ben Yaacov’s Grothendieck note identifies this double-limit property with weak almost periodicity/relative weak compactness of the corresponding family of functions.

The packet does not prove Conjecture 1. The strongest failed route in this push was to combine the Dvoretzky–Milman finite-dimensional subspaces with definability of stable types. That yields an almost-monochromatic Hilbert subspace in an elementary extension or ultrapower, but it does not by itself produce an actual infinite-dimensional subspace of the original Y . The WAP midpoint lemma is exactly the missing finite-to-infinite pullback step.

Novelty and Literature Check

The push checked Hanson [1], Ben Yaacov [2], Shelah–Usvyatsov [3], Chaatit [4], Braga–Swift [5], and Gowers-style oscillation-stability literature. No existing full answer to Hanson Question 4.3 was found. The closest known tools are the Grothendieck/WAP characterization of stable formulas and the Hilbert Morley-spine machinery for minimal wide stable types. Neither currently supplies the in-model subspace required by oscillation stability.

Human Review Recommendation

Review Conjecture 1. A proof would promote this packet to a full solution of Hanson Question 4.3. A counterexample would give an analytic template for trying to build a stable Banach-theoretic counterexample, although additional work would still be needed to realize the midpoint-WAP function as a formula in a stable Banach theory.

References

- [1] James Hanson. *Indiscernible Subspaces and Minimal Wide Types*. arXiv:2004.03062, 2020.
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- [3] Saharon Shelah and Alexander Usvyatsov. *Minimal types in stable Banach spaces*. Advances in Mathematics 355 (2019), arXiv:1402.6513.
- [4] Fouad Chaatit. *A Representation of Stable Banach Spaces*. arXiv:math/9602208, 1996.
- [5] Bruno de Mendonca Braga and Andrew Swift. *Coarse embeddings into superstable spaces*. arXiv:1704.04468, 2017.